

Kalin Kochnev

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EDUCATION

University of Connecticut

Bachelor of Science and Engineering, Robotics Engineering

Storrs, CT

Aug 2021 — Dec 2025

Cumulative GPA: 3.972/4.0 | Dean's List Fall 2021 - Spring 2025, Honors Program, STEM Scholar, NASA Space Grant

TECHNICAL SKILLS

Robotics: ROS, Motion Planning (APF/A*/PRM/RRT and variants), Kinematics/Dynamics Modeling, Lyapunov/LTI Stability Analysis, BLDC Motor Control, Sensing, Numerical Algorithms, UR3 Robot Arm, Machine Learning.

Mechanical: OnShape, Inventor/Vault Professional, Additive Manufacturing, MIG Welding, Waterjet, Tabletop CNC.

Programming: STM32/ESP32/AVR128DB Firmware, FreeRTOS, CMSIS-RTOS, JTAG, Hardware-in-Loop Testing, Rust, C/C++, Python, Matlab, WebGPU, Fullstack Web Development, Linux, GDB, Git, Docker.

Electrical: Battery Management, 3-Phase Power, Sensing, I2C/CAN/Serial, Motor Control, Digital Signal Processing

WORK EXPERIENCE

Robotics Engineer

Oct 2025 — May 2026

Reefgen

San Francisco, CA

- Containerized Gazebo simulation environment for portability across developers computers.
- Designed and implemented a service in ROS to plan and orchestrate groups of actions to execute a mission plan.
- Documented and updated all electrical schematics into a unified hierarchical representation in KiCAD for inter-assembly connections as well as per-board connections.

Undergraduate Researcher

Sep 2023 — March 2026

Fluids, Reduction, Nonlinearity, and Turbulence Lab — University of Connecticut

Storrs, CT

- Applied Lyapunov stability analysis and convex optimization to analyze stability of time-dependent shear flows.
- Secured a \$4k research grant and a \$3k NASA Space Grant scholarship.
- [Manuscript](#) was accepted 2026 American Control Conference.

Robotics Engineering Intern

Jun 2025 — Aug 2025

Good Machine/Reefgen/Pelagic

San Francisco, CA

- Troubleshooted and produced detailed documentation of failing battery management system for underwater ROV.
- Developed modular and parametric CAD design for rapid release of underwater ROV thruster filters in the field. Passed lab and in-field testing.
- Ported accelerometer, magnetometer, power management, global datetime handling functionality from an obsolete MCU architecture to a STM32/CMSIS-RTOS based device for low power sea vessel tracking.

Undergraduate Researcher

Sep 2024 — May 2025

Intelligent Systems and Control Laboratory — University of Connecticut

Storrs, CT

- Researched and implemented in *roscpp/rospy* cutting edge motion planning algorithms for cooperating UR3 robot manipulators for additive manufacturing as a senior capstone project ([video](#)).

Embedded Engineering Intern

Jun 2024 — Aug 2024

Gilbarco Veeder Root

Simsbury, CT

- Developed sensor drivers and data collection utilities for the STM32 to target new high value markets.
- Saved the company \$6K by identifying cost-effective cable substitutes.
- Diagnosed and resolved a rare field issue, saving technicians and customers significant time.

Software and Hardware Team Lead

Sep 2021 — Nov 2022

2022 NASA BIG Idea Challenge — University of Connecticut

Storrs, CT

- Led an undergraduate team funded \$150K by NASA to design a quadruped lunar vehicle.
- Built CAN protocol in Rust for brushless motors/encoders with GUI dashboard and ROS interface

Projects (visit <https://kalinkochnev.com/journey>)

- Self-piloting Bike, Multi-arm Collaborative 3D Printing, Stability Analysis of Time-Varying Shear Flows.